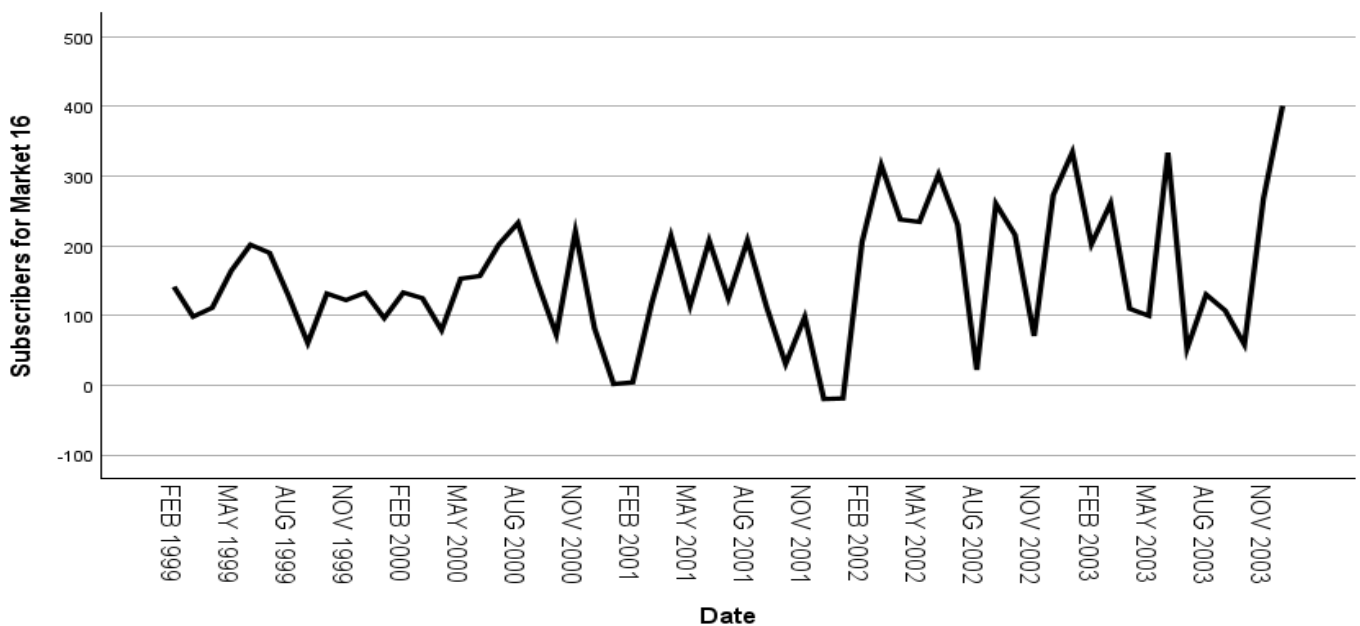
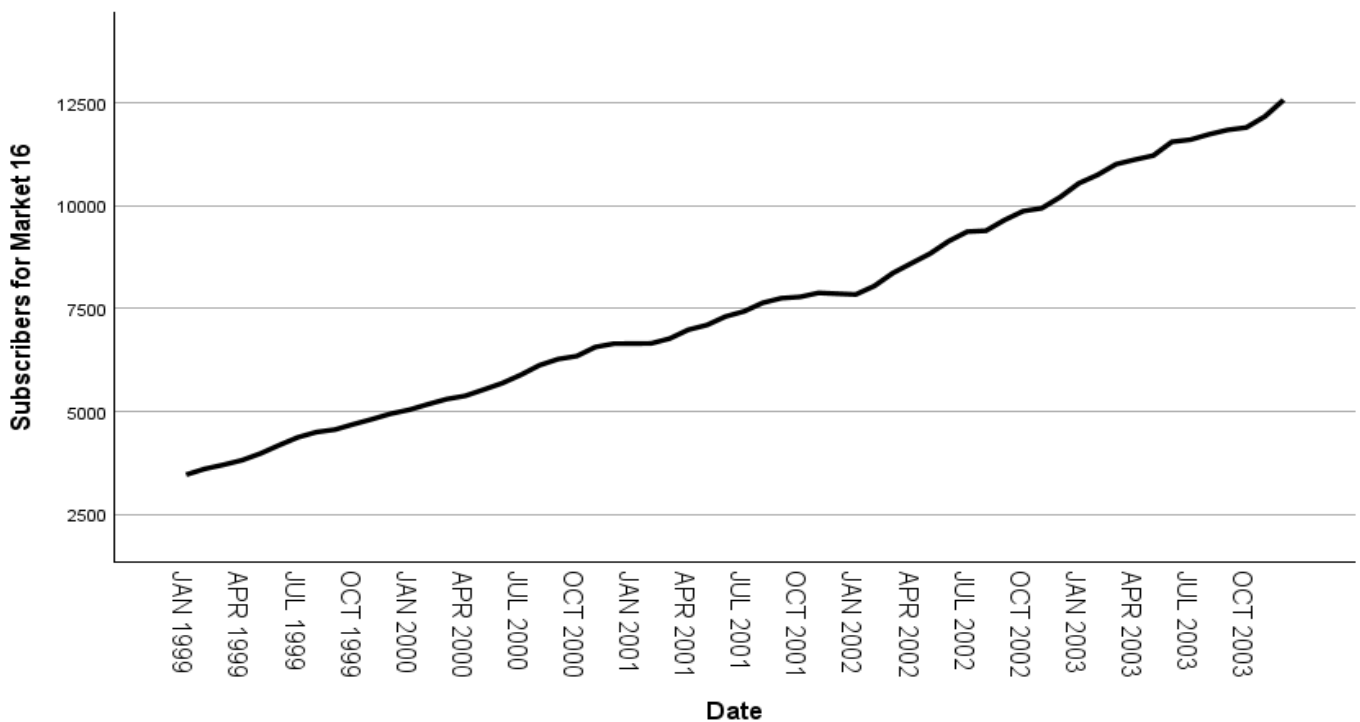


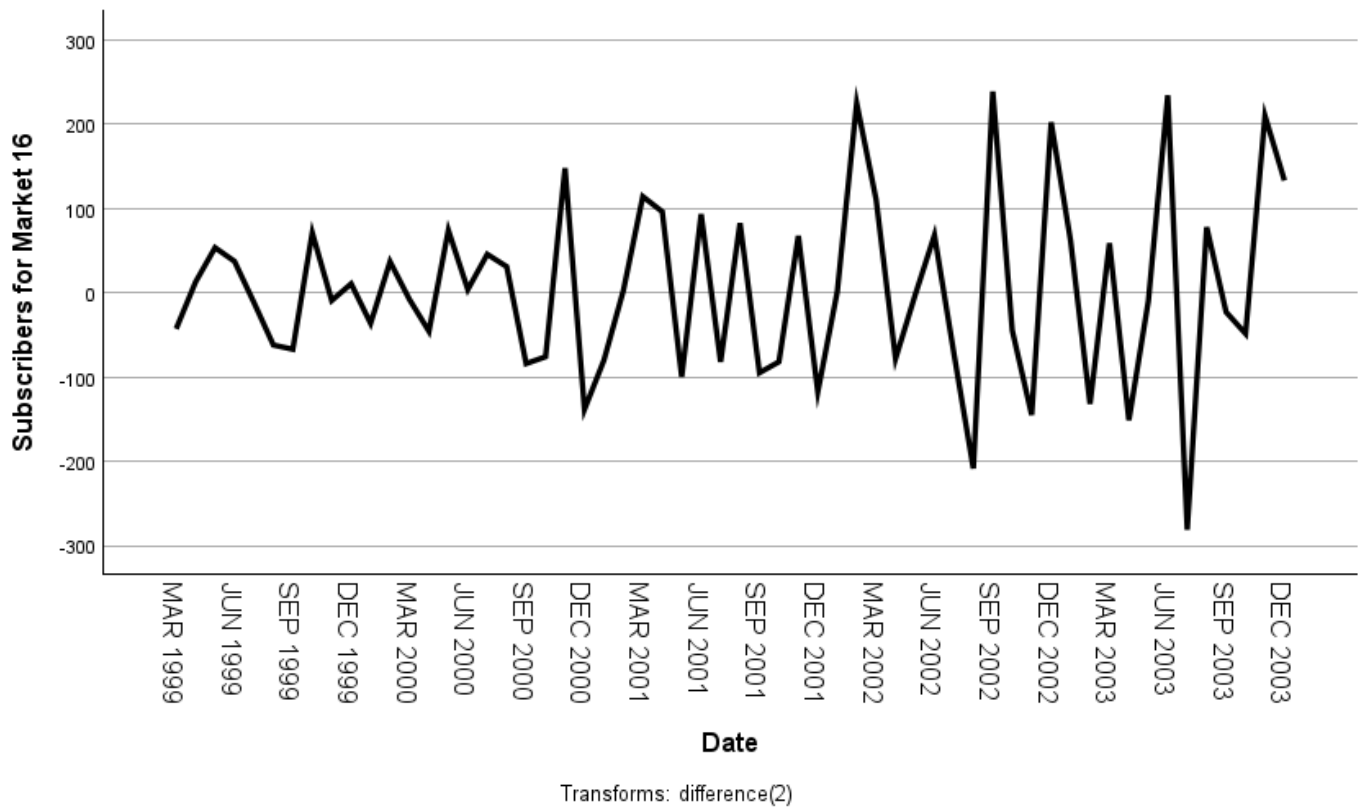
Sequence Chart Analysis

For this exercise, I selected Subscribers for Market 16 from the broadband dataset. The original sequence chart showed a strong upward trend in subscriber counts over time.

To remove the trend, differencing was applied. The first-order differenced sequence chart showed the month-to-month changes in subscriber counts, but some trend and variability remained. A second-order differenced sequence chart was then created. The second-order differenced series fluctuated around zero and appeared much more stationary. Based on visual inspection of the differenced sequence charts, second-order differencing was selected for the final ARIMA model.



Transforms: difference(1)



ARIMA Model Development

Rather than using the Expert Modeler, I manually developed an ARIMA model based on the characteristics of the time series and residual diagnostics.

After examining the sequence charts, I determined that second-order differencing was necessary to remove the trend and create a stationary series. I then specified an ARIMA(1,2,1) model.

Model Description

			Model Type
Model ID	Subscribers for Market 16	Model_1	ARIMA(1,2,1)(0,0,0)

The components of the model are:

AR(1): One autoregressive component that uses the previous period's value to predict the current period.

I(2): Second-order differencing used to remove the trend and create stationarity.

MA(1): One moving average component that uses the previous forecast error to improve future predictions.

The estimated model parameters were:

$$AR(1) = 0.285$$

$$MA(1) = 0.996$$

These parameters indicate that both previous values and previous forecast errors contribute to forecasting subscriber growth.

ARIMA Model Parameters

				Estimate	SE	t	Sig.
Subscribers for Market 16-Model_1	Subscribers for Market 16	No Transformation	Constant	1.594	1.066	1.495	.141
			AR Lag 1	.285	.160	1.782	.080
			Difference	2			
			MA Lag 1	.996	1.019	.978	.333

Model Fit Statistics

The model produced a Stationary R-Squared of 0.333. This means that approximately 33.3% of the variation in the stationary series is explained by the model. Higher values indicate a better-fitting model.

The Mean Absolute Percentage Error (MAPE) was 0.925%. This means that the average forecasting error was less than 1%, indicating excellent predictive accuracy.

The Maximum Absolute Percentage Error (MaxAPE) was 2.046%. This means that even the largest forecasting error observed in the model was only about 2%, which indicates strong forecasting performance.

Model Statistics

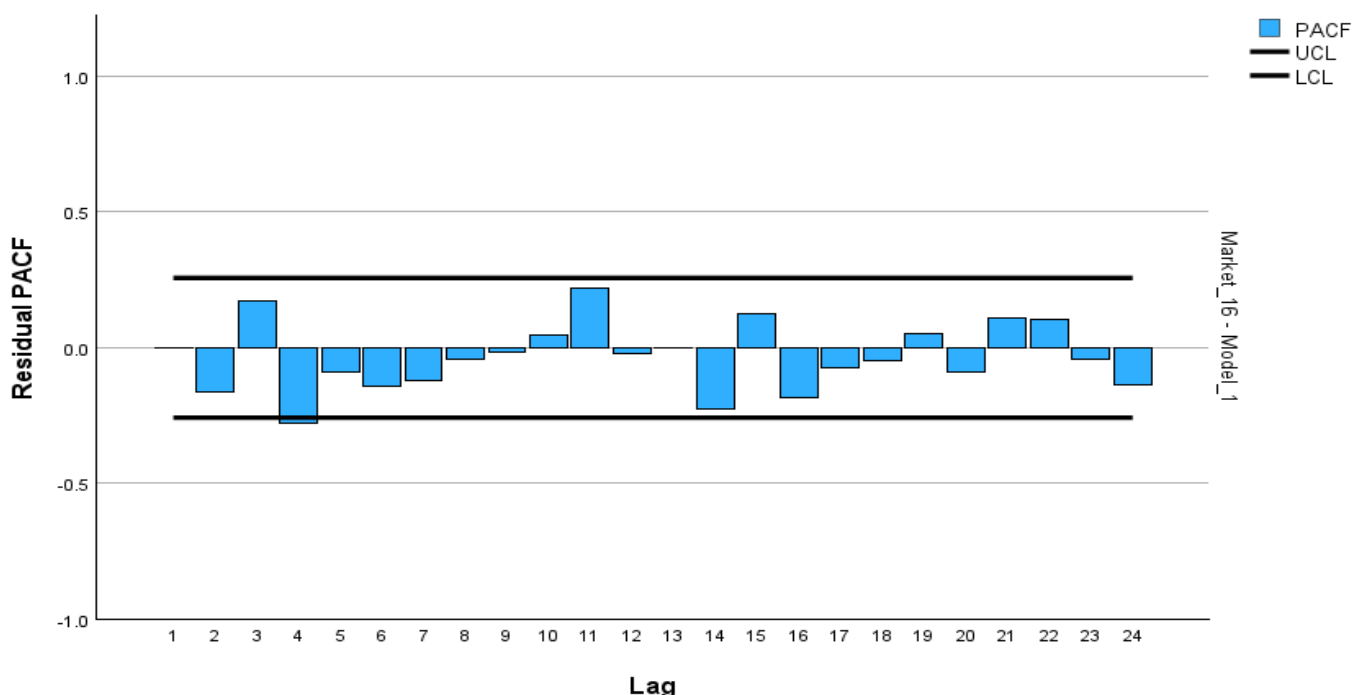
Model	Number of Predictors	Model Fit statistics			Ljung-Box Q(18)			Number of Outliers
		Stationary R-squared	MAPE	MaxAPE	Statistics	DF	Sig.	
Subscribers for Market 16-Model_1	0	.333	.925	2.046	22.084	16	.141	<.001

Overall, the model demonstrated a good fit and accurate predictions.

Assessment of Autocorrelation

The residual PACF plot was examined to evaluate whether significant autocorrelation remained in the residuals.

Most lag values fell within the upper and lower confidence limits. Although one lag showed a minor spike, it was only slightly outside the confidence interval and did not indicate a substantial autocorrelation problem.



The Ljung-Box test produced the following results:

Ljung-Box Q(18) = 22.084

Significance (p-value) = 0.141

Because the significance value is greater than 0.05, the null hypothesis of no residual autocorrelation cannot be rejected.

Model Statistics								
Model	Number of Predictors	Model Fit statistics			Ljung-Box Q(18)			Number of Outliers
		Stationary R-squared	MAPE	MaxAPE	Statistics	DF	Sig.	
Subscribers for Market 16-Model_1	0	.333	.925	2.046	22.084	16	.141	<.001

Therefore, the residuals appear independent, and the model satisfies the assumption of no significant autocorrelation.

Forecasted Values

The ARIMA(1,2,1) model was used to generate forecasts beyond the build sample period.

The forecast indicated continued growth in subscribers for Market 16. Forecasted values followed the historical upward trend observed in the original data.

The forecast confidence intervals remained relatively narrow, suggesting that the model produced stable and reliable forecasts.

